

Multiple Choice

1. **Answer:** B

Options: A) The integers under addition; B) The nonzero integers under multiplication; C) The nonzero real numbers under multiplication; D) The complex numbers under addition

Note: Correct option: B - The nonzero integers under multiplication

2. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 4

Note: Correct option: D - 4

3. **Answer:** C

Options: A) $1/2$; B) $7/12$; C) $5/8$; D) $2/3$

Note: Correct option: C - $5/8$

4. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

5. **Answer:** C

Options: A) $2/69$; B) $1/30$; C) $2/23$; D) $12/125$

Note: Correct option: C - $2/23$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'both symmetric and anti-symmetric'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3'.

Long Form Response

11. **Answer:** Sum the first two equations to find $c + o + n = t$. Solve the third equation for c to find $c = s - t$, and substitute $s - t$ for c in $c + o + n = t$ to find $o + n + s - t = t$ or $o + n + s = 2t$. Substitute 12 for $o + n + s$ to find $t = 12/2 = 6$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** $S(n)$ is not true for any $n \leq 0$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key